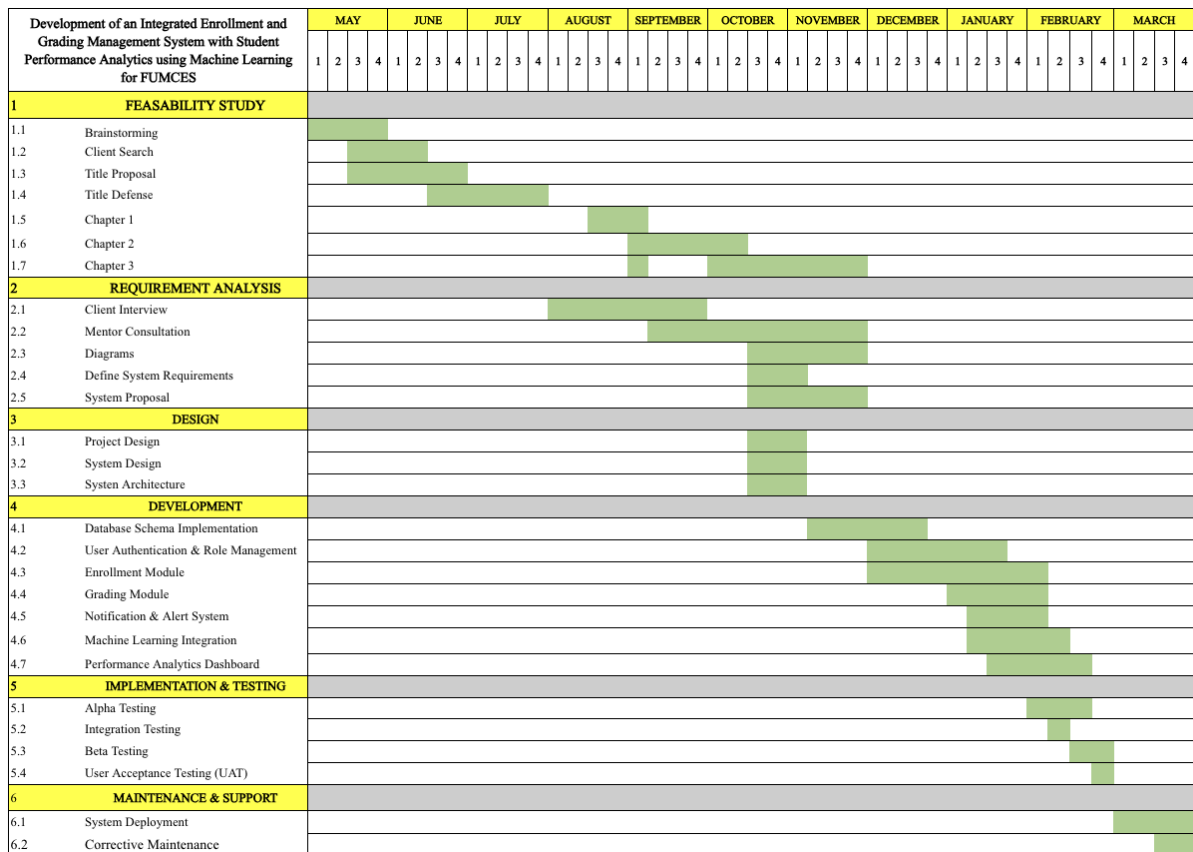




**Figure 14: Project Gantt Chart**

The Development of an Integrated Enrollment and Grading Management System with Student Performance Analytics using Machine Learning for FUMCES is driven by an extensive project timeline, which is set between May 2025 and March 2026. The Gantt chart also shows the project broken down into six key stages namely, Analysis, Design, Development, Testing, Deployment, and Documentation in such a way that the system development process is structured and goal-oriented.

The Analysis Phase will be between May 2025 and August 2025. At this stage, there is preliminary brainstorming, title proposal, and the development of the system's conceptual framework and technical background. These preliminary activities determine the course and extent of the system. The key activities of requirement gathering and system flow

development are performed to determine the main system requirements, ending with a complete architectural proposal.

The Design Phase, which is conducted from August 2025 to November 2025, is concerned with technical planning such as database design, user interface prototyping, and the creation of architectural frameworks that will steer implementation. The Development Phase starts in November 2025 and ends in February 2026. The functional elements like user management, enrollment processing, sectioning, scheduling, grading modules, and the integration of performance analytics are developed at this stage to provide a sound system.

The Testing Phase takes place in February and March 2026 and includes unit testing, integration testing, and user acceptance testing to ensure the system satisfies institutional expectations. The Deployment Phase occurs in March 2026, focusing on the actual implementation of the system and user training. Lastly, the Documentation Phase runs concurrently with the development and final stages, concluding in March 2026. This final stage guarantees that all system manuals and research findings are finalized for long-term usability.

The sequential flow of phases shown in the Gantt chart allows to track the progress on a regular basis and use resources efficiently, which helps to successfully and timely implement the FUMCES system.